

Skill Recorder Evaluation — Cold vs Oracle

10 个 WebArena shopping_admin 任务, Claude Code CLI 驱动, seed 0, sonnet 4.6

报告日期	2026-05-26
Runs	Cold 20260525T084638Z · Oracle 20260526T040953Z
Benchmark	WebArena shopping_admin only (单服务 MVP scope)
Arms tested	cold · oracle
Seeds	0 (single seed; Wilson CI is wide — see below)

Executive Summary

Oracle SKILL.md takes Claude Code from cold 60.0% → 100.0%, **6.2× cheaper**, **3.5× fewer turns**.

† One cold trial uses a manual override of the original CSV score (webarena.704 — original 0 was a runner_ccc pw.Page.url cache bug; manual re-run after the fix confirmed 1.0).

Summary by arm

Metric	Cold	Oracle
Success rate	60.0%	100.0%
95% Wilson CI	[31.3%, 83.2%]	[72.2%, 100.0%]
Trials (real)	10	10
Total cost	\$6.18	\$0.99
Avg cost / trial	\$0.62	\$0.10
Total turns	308	88
Avg turns / trial	30.8	8.8
Total wall-clock	29.8 min	7.8 min

Per-task results

每格:  /  / — (未运行或 † 主动跳过) + turns + cost。† 表示覆盖了原始 CSV 分数。

Task	Cold	Oracle
webarena.453	✓ 47t / \$0.63	✓ 9t / \$0.10
webarena.458	✓ 7t / \$0.13	✓ 8t / \$0.11
webarena.470	✓ 8t / \$0.14	✓ 7t / \$0.08
webarena.496	✓ 24t / \$0.55	✓ 9t / \$0.11
webarena.538	✓ 51t / \$1.34	✓ 11t / \$0.12
webarena.771	✗ 30t / \$0.63	✓ 12t / \$0.12
webarena.423	✗ 51t / \$0.85	✓ 6t / \$0.07
webarena.704	✓ † 8t / \$0.14	✓ 6t / \$0.07
webarena.694	✗ 31t / \$0.67	✓ 11t / \$0.12
webarena.699	✗ 51t / \$1.09	✓ 9t / \$0.10

Five product-level findings (surfaced by oracle authoring)

All five are real Skill Recorder TODOs, encoded as work-arounds in the oracle SKILL.md "Critical tool notes" sections. Fixing them should narrow the human → oracle gap.

#	Issue	Impact	Current work-around	Product fix direction
1	<code>browse</code> <code>fill</code> presses Enter by default	Form fields submit prematurely, breaking subsequent fills	SKILL.md adds <code>--no--press-enter</code>	Renderer default to <code>--no--press-enter</code> ; only add Enter for <code>action: "submit"</code>
2	<code>browse</code> <code>fill</code> doesn't work on <code><select></code>	Dropdown values silently not applied — biggest single cause of human-arm failures	Hand-written oracle uses <code>browse select</code>	Add <code>action: "select"</code> to Skill schema; renderer outputs <code>browse select</code> for selects
3	<code>browse</code> <code>fill</code> doesn't work on Magento checkbox toggles	Toggle widgets (Enable Product, Sale, etc.) cannot be flipped	Click adjacent <code><label></code> (<code>input[name="..."] + label</code>)	Renderer detects checkbox and emits click-label pattern
4	Playwright <code>Page.url</code> cache stale after external CDP nav	URLEvaluator reads stale URL after CCC's <code>browse open</code> , false-fails <code>url_match</code>	<code>runner_ccc</code> calls <code>page.evaluate(...)</code> before validate to force a CDP roundtrip	Use <code>frame.url()</code> or fresh CDP target enumeration in evaluator
5	Claude Code doesn't auto-load project skill bodies	SKILL.md name+description seen in catalog, body ignored — agent improvises and gets details wrong	runner prompt nudges "read your skills before acting"	Document skill-discovery patterns; possibly hint built-in

Methodology — how the three SKILL.md sets were produced

- **Cold:** no SKILL.md. The agent receives only the WebArena task intent and the `browse` tool surface. Tests Claude Code's first-encounter capability.
- **Oracle:** 10 hand-authored SKILL.md files following our four-section template (Background, Critical tool notes, Steps, On failure). 4 were authored by direct Magento exploration (704, 771,

423, 694, 699); 5 were produced by parallel sub agents (453, 458, 470, 496, 538), each given a self-contained prompt with the task intent + WebArena eval criteria + Magento credentials + the five known browse-CLI gotchas + a format reference. All sub-agent skills passed eval on first try.

Next steps

1. **Fix Findings #2 and #3 first** (the `<select>` and checkbox cases). These are the single biggest causes of human-arm failures, and they're well-scoped schema + renderer changes — no behavioural changes to CRX recording.
2. **3 seeds × 3 arms** (\$20-25, ~4h): tightens Wilson CI from $\pm 30\text{pp}$ to $\pm 15\text{pp}$; closes the door on "this is a single-seed fluke" objections.
3. **opus 4.7 robustness check** (\$10): confirm conclusions generalise across frontier models.